Exercise 2.
SQL fundamentals (Aggregate functions & Grouping)

1. Write a SQL query to retrieve all columns from the employees

```
SELECT Count(*) As total_employees
FROM employees
```

total_employees
10

2. Write a SQL query to find all the unique departments in the employees table

```
SELECT Sum(Salary) As total_salary_IT
FROM employees
WHERE department 'IT';
```

total_salary_it
220000

3. Write a SQL query to retrieve all employees first and last names, ordered by salary in descending order

```
SELECT Av(Salary) As Avg_hr_salary
FROM employees
WHERE department = 'HR';
```

avg_hr_salary
49500

4. Write a SQL query to retrieve the top 5 highest-paid employees

~~SELECT MIN(salary) AS lowest_salary.~~

SELECT MIN(salary) AS lowest_salary, MAX(salary AS highest_salary
FROM employees;

lowest_salary	highest_salary
48000	62000

~~5. Write a SQL query to find employees who work in the IT dep~~

5. Write a SQL query to group employees by department and display the total salary paid in each department

SELECT department SUM(salary) AS total_salary
FROM employees
GROUP BY department;

department	total_salary
IT	220000
HR	99 000
Finance	119000
Marketing	105 000

6. Write a SQL query to count how many employees work in each city

SQL query:

SELECT city
COUNT(*) AS employee_count
FROM employees
GROUP BY city;

City	employee_count
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. Question: Write a SQL query to group employees by department, calculate the average salary in each department and order the results in descending order of average salary.

SQL query.

```
SELECT department,
       AVG(salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC;
```

Department	avg_salary
IT	55000
finance	59500
HR	49500
marketing	52500

8. Question: Write a SQL query to find departments where the total salary paid exceeds 100,000.

SQL query:

```
SELECT department,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100000
```

department	total_salary
IT	220000
finance	119000
marketing	105000

9. Question: Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

SQL query:

```
SELECT city,
       COUNT(*) AS employee_count
FROM employees
GROUP BY city
HAVING COUNT(*) > 1
ORDER BY employee_count DESC;
```

City	employee_count
Chicago	3
New York	2
Los Angeles	2
San francisco	2

10. Question: Write a SQL query to find the department with the highest average salary

SQL query

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC
LIMIT 1;
```

Department	avg_salary
IT	55000